

Acceleration and variability in vocabulary growth for young children and language models

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AI disclosure

- First project I've collaborated completely with AI (Claude Code)
- ~6 weeks of constant iteration
- Being careful about requesting cleanup, documentation, best practices throughout
- Question about how to establish trust?
 - Key plots
 - Annotation of Stan code



Vocabulary growth

Theoretical: Vocabulary is a key nexus of language, social cognition, and conceptual development.

Practical: Variation in vocabulary growth is a key precursor of academic outcomes.

Methodological: Vocabulary count has a property nearly no other psychological metric has: a true zero, so we can explore the psychometrics of learning.

Comparative: We can explore how LLMs relate to children using dense data about child outcomes.

Wordbank Data

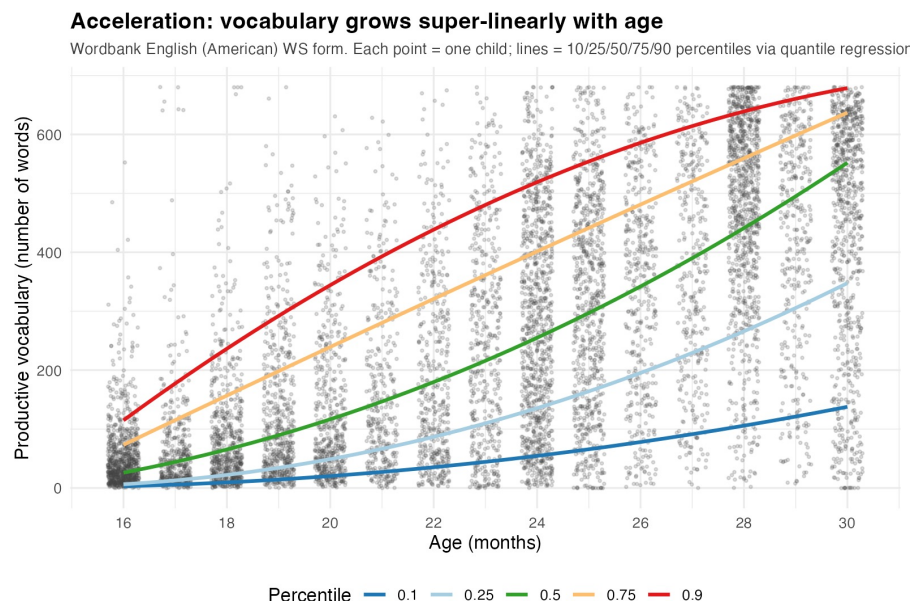
MacArthur–Bates Communicative Development Inventory (CDI): parent-report checklist of ~ 680 words.

We focus on production (less noisy).

Wordbank: open repository of CDI administrations. ~ 30 languages, ~ 10^5 children. We use both:

- **Cross-sectional:** one snapshot per child
- **Longitudinal:** ≥ 2 admins per child, lets us track per-child growth

Large N , parent-report calibrated to direct testing, cross-linguistically validated.

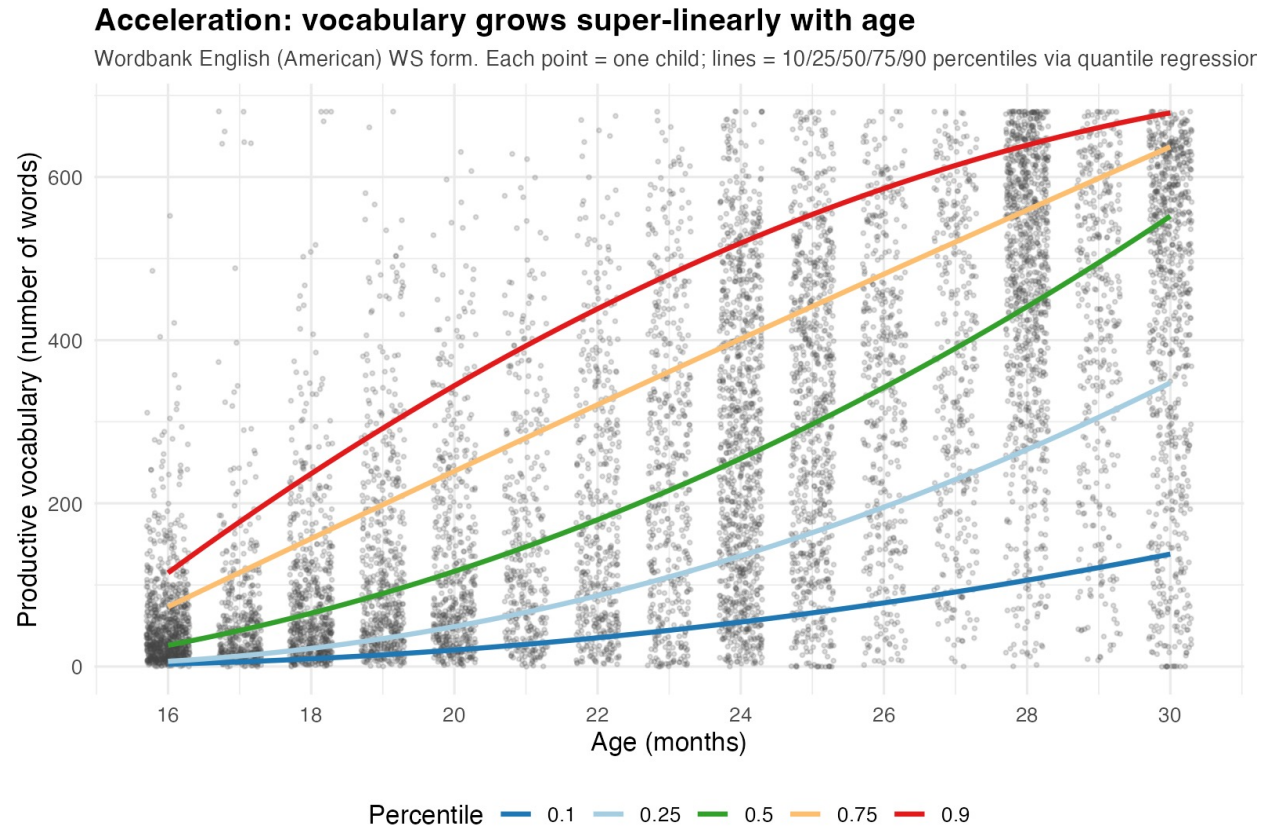


Two signatures of vocabulary growth

Acceleration +
variability are
ubiquitous

Terminology note:
Efficiency: what's
changing in the child

Acceleration: what's
happening in the data



Is growth pure accumulation?

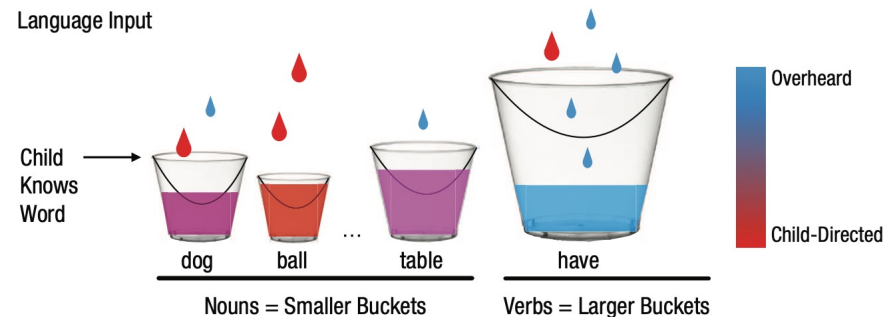
Maybe acceleration just falls out of accumulation?

Each child has input rate r_i ; word j has difficulty δ_j . A word is acquired when enough tokens accumulate.

Derived from cross-situational word learning and other associative/statistical learning ideas.

Acceleration: Some models have changes in efficiency to create acceleration, others do not.

Variation: Between-child variation only in baseline (r_i or α_i). No variation in efficiency.



In the literature:

- McMurray (2007) — pure rate $\times$ time
- Mitchell & McMurray (2009) — + leverage feedback
- Hidaka (2013) — cumulative-Weibull; rate change
- Mollica & Piantadosi (2017) — per-word Gamma
- Kachergis et al. (2021) — IRT Rasch + α_i

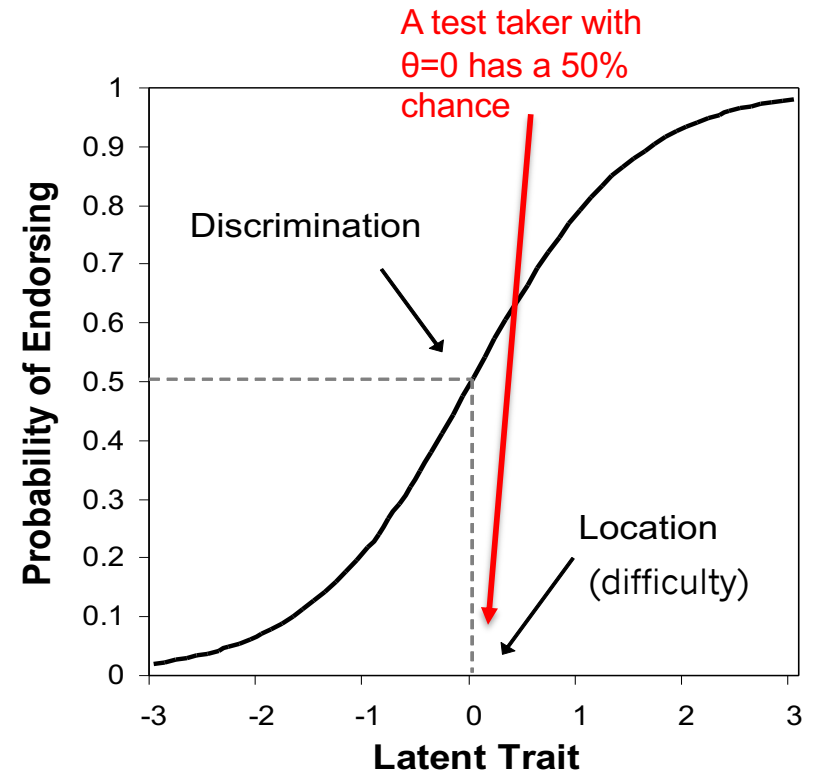
Item response theory (IRT)

IRT describes a family of models of binary data (like CDI data).

We use the Rasch model, a logistic model where each item has a single difficulty parameter δ , and each person has a single ability parameter θ .

Probability of success on an item is then just:

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$



ICC = item characteristic curve

Puzzle: input matters less than supposed

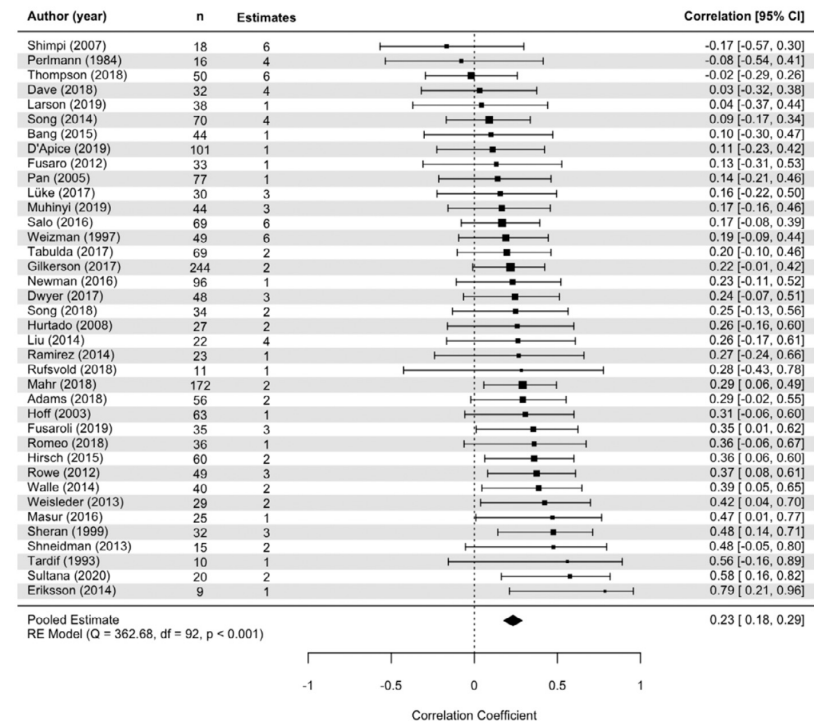
The accumulator view predicts variation in growth comes from variation in input quantity.

Coffey & Snedeker (2026) meta-analyzed 71 studies, 4,760 children:

R^2 of input on outcome = 0.04–0.07.

Caregiver speech quantity explains 4–7% of variance in language outcome — a real but small effect.

Hypothesis: Variability comes from *how children accelerate*, not how much input they hear.



This work: modeling individual variability

We build a Bayesian model to estimate three things directly:

1. **Efficiency gain.** Does acceleration result from changes in learning efficiency?
2. **Variability.** Do children differ in *how* and *when* their growth accelerates, not just in baseline ability?
3. **Input share.** How much of the between-child variance is attributable to input rate, vs. to internal factors like variation in efficiency?

Does the same machinery applied to large language models yield the same answers?

Spoiler: no — LLMs show a categorical disanalogy with children's -learning.

BUILDING THE MODEL

The accumulator hypothesis

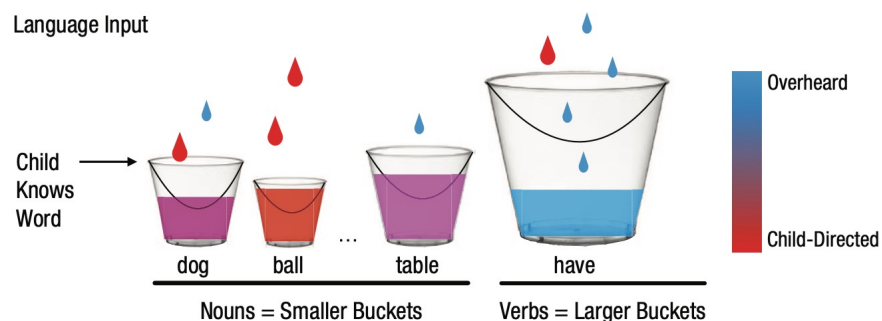
The idea. Each child accumulates word-evidence at her own input rate r_i ; word j is acquired once its difficulty δ_j is crossed.

Ability scales as a power of input. If kids vary only in *rate*, then ability grows as (tokens)¹ — the pure *unit accumulator*.

Between-child differences come only from differences in r_i .

Model checks:

- Is the growth exponent really 1, or steeper?
- Are children all alike except for input rate, or do they also differ in efficiency or in growth rate?



Rasch IRT

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}.$$



Standard Rasch IRT model – the substantive content is in how θ_i depends on accumulated experience.

Rasch IRT + accumulator dynamics

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}, \text{ where:}$$

$$\theta_i = \log r_i + \log t$$


per-child input rate *time = age*
(words per month) *(months since onset)*

Ability = rate × time. The only thing that changes across development is more accumulated input. The only source of variation between children is variation in the input rate

Rasch accumulator + per-child efficiency

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}, \text{ where:}$$

$$\theta_i = \underset{\substack{\uparrow \\ \text{input rate}}}{\log r_i} + \underset{\substack{\uparrow \\ \text{per-child} \\ \text{efficiency} \\ \text{(NEW)}}}{\log \alpha_i} + \underset{\substack{\uparrow \\ \text{time}}}{\log t}$$

Each child has a multiplicative efficiency α_i over and above their input rate. **The question we'll ask of the fit:** how much of the between-child variance comes from efficiency vs. input rate? We summarize this share as $\pi_\alpha = \sigma_\alpha^2 / (\sigma_\alpha^2 + \sigma_r^2)$.

Rasch accumulator + efficiency slope

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}, \text{ where:}$$

$$\theta_i = \underset{\substack{\uparrow \\ \text{input rate}}}{\log r_i} + \underset{\substack{\uparrow \\ \text{efficiency}}}{\log \alpha_i} + \underset{\substack{\uparrow \\ \text{population} \\ \text{scaling exponent} \\ \text{(NEW; shared)}}}{\kappa_{\text{pop}}} \cdot \underset{\substack{\uparrow \\ \text{time}}}{\log t}$$

The population accelerates: ability scales as $t^{\kappa_{\text{pop}}}$ super-linearly. All children still share one κ_{pop} . **What we'll ask:** is κ_{pop} really above 1 — i.e., does ability grow faster than a unit accumulator predicts? This is Hidaka's rate-change model.

Rasch accumulator + per-child efficiency slope

$$P(w_{i,j} = 1 | \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}, \text{ where:}$$

$$\theta_i = \log r_i + \log \alpha_i + (\kappa_{\text{pop}} + \zeta_i) \cdot \log t$$



efficiency pop. scaling *per-child
acceleration
deviation
(NEW)*

Each child has their own growth exponent κ_i
 $= \kappa_{\text{pop}} + \zeta_i$. The new piece: between-child
variation isn't just in baseline ability, but in
how trajectories accelerate. Prior accumulator
models force every child to share one growth
rate.

RESULTS: ENGLISH AND NORWEGIAN LONGITUDINAL FITS

Data and compute for the longitudinal fits

Data.

- **Reduced English (WS form).** $I = 200$ children, $J = 200$ words.
- **English.** $I = 500$ children, ~ 1500 administrations, $J = 671$ items (full WS list), $N \approx 1.1\text{M}$ observed responses. Stratified-by-median-age sampling.
- **Norwegian.** $I = 500$ children, ~ 3000 administrations, $J = 716$ items, $N \approx 2.2\text{M}$ responses. Same stratification.

Model family. Four-step build run as a sequence of Stan models on each dataset: `pure`, `+ $\alpha$` , `+ $\kappa$ pop`, `+ $\zeta_i$` .

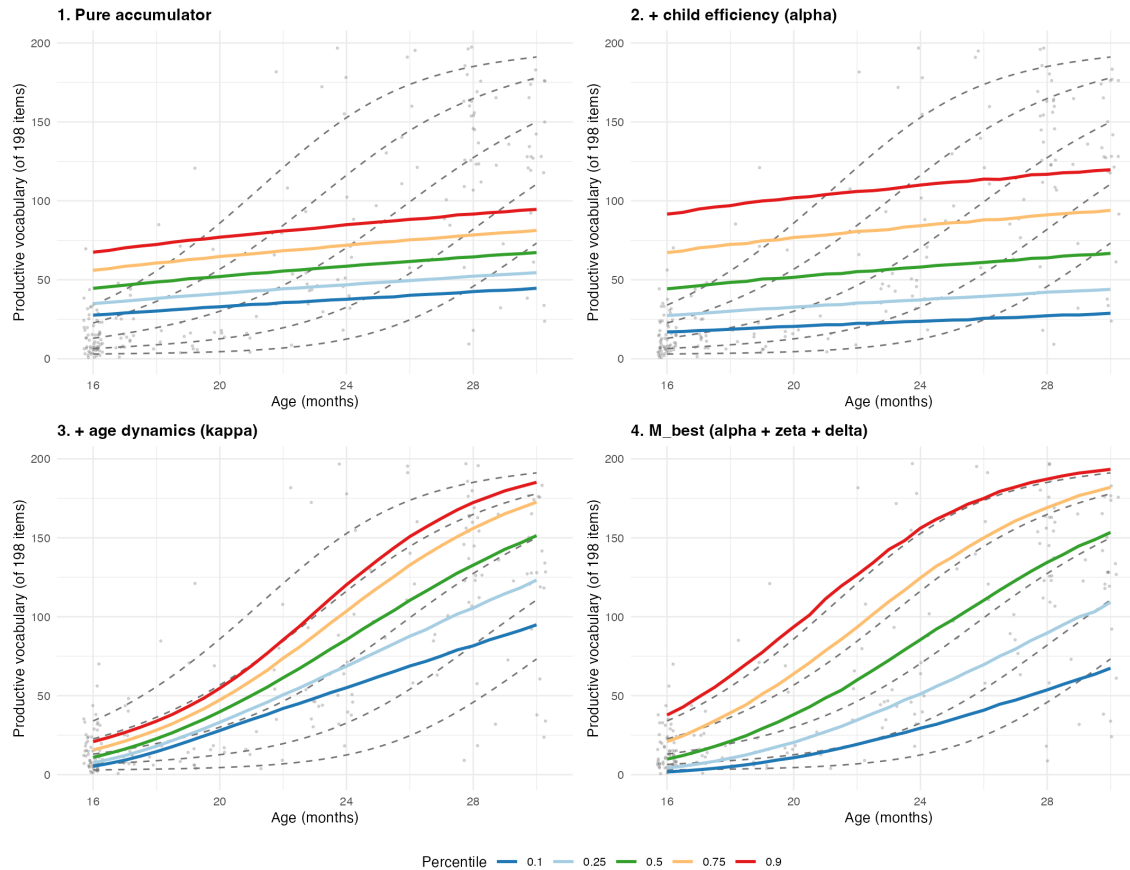
Compute.

- Sherlock for reduced fits.
- GCP virtual machine (`n2d-standard-128`, 64 physical AMD EPYC cores each, 128 GB RAM). Stan with `reduce_sum` threading: 4 chains $\times$ 16 threads/chain saturates the 64 physical cores per machine.

Comparison to data

Building up M_best (English I=200, J=198)

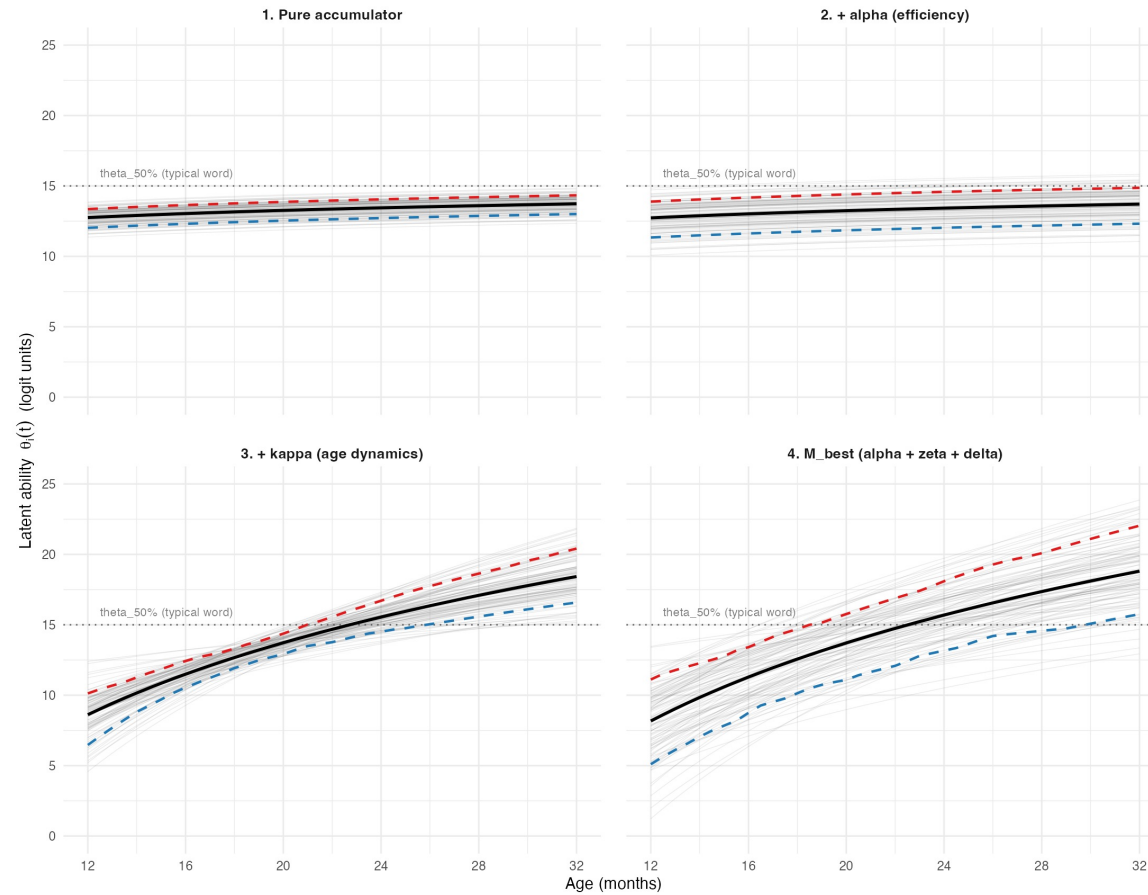
Each panel = one component added. Empirical: grey dots (1 admin/kid x-sec) + dashed lines (GAMLSS BEINF lambda=10000).



Latent ability changes

Building up M_{best} ($l=200$, theta-space view)

100 simulated kids/panel. Grey: per-kid trajectories. Black: pop mean. Blue/red dashed: 10/90th percentiles. Dotted: theta where typical word reaches 50%.



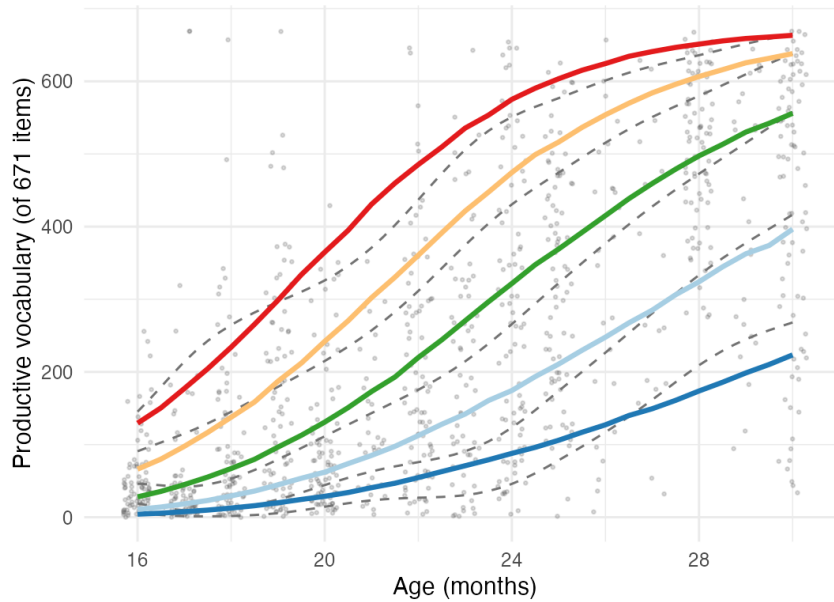
N=500 longitudinal data

M_best vs. wordbank-wide cross-sectional empirical

Model: longitudinal l=500 fit ($\alpha + \zeta + \delta$, no s, no s_i). Empirical: full wordbank WS, one admin/kid (random), GAMLSS BEINF $\lambda=10000$.

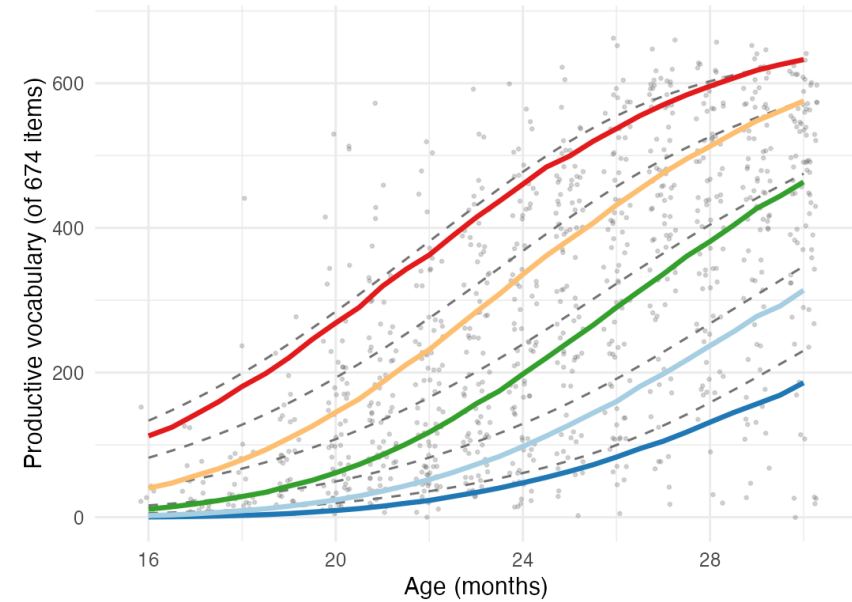
English

1636 kids (1 admin/kid; wordbank-wide x-sec)



Norwegian

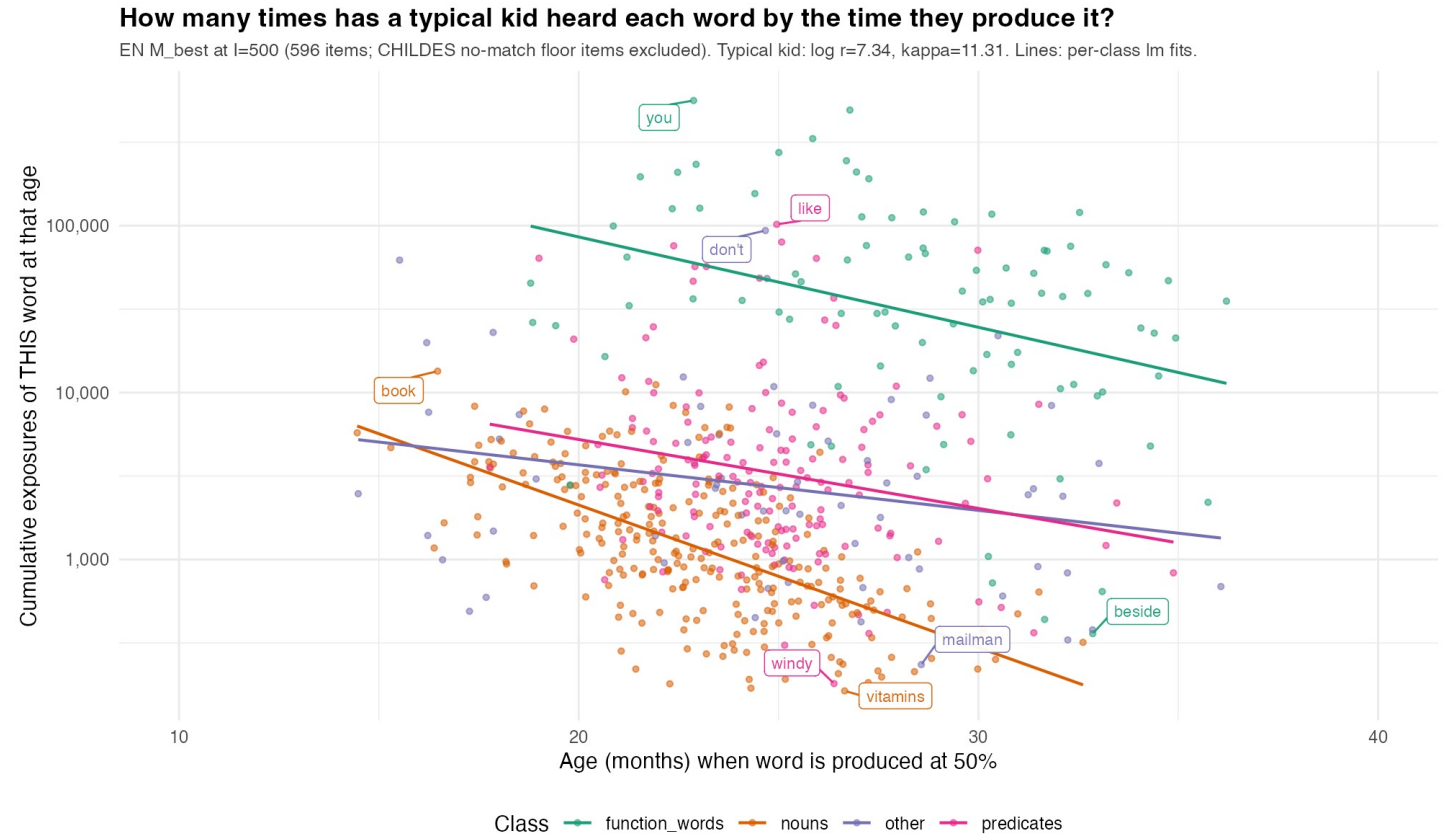
1473 kids (1 admin/kid; wordbank-wide x-sec)



Percentile — 0.1 — 0.25 — 0.5 — 0.75 — 0.9

Exponentially fewer exposures before learning

- d



Best-fitting model: parameter posteriors

Parameter	EN M_best (I=500)	NO M_best (I=500)
I (kids)	500	500
J (items)	671	674
N (obs)	1106805	2056872
delta	10.31 [10.25, 10.37]	11.47 [11.40, 11.53]
sigma_alpha	1.81 [1.69, 1.94]	2.05 [1.93, 2.18]
sigma_zeta	3.81 [3.58, 4.07]	4.79 [4.49, 5.13]
pi_alpha	0.92 [0.91, 0.93]	0.94 [0.93, 0.94]
rho(xi, zeta)	-0.09 [-0.18, -0.01]	-0.13 [-0.21, -0.04]

Cross-language acceleration

$\kappa_{\text{pop}} \sim 10 - 12$ **in both languages**, with $P(\kappa_i > 1) > 0.997$.

δ **vs. ζ_i decomposition is identifiability-dependent.** In Norwegian (8 admins/kid), per-child slopes are well-identified — dropping ζ_i *hurts* fit by 235 ELPD. In English (3 admins/kid), δ pools what Norwegian's data lets fall into ζ_i .

Reading. The acceleration signature is universal. Its internal parameterization is identifiability-dependent on data density.

In any dense longitudinal sample, between-child κ_i spans roughly 5 – 15.

INPUT QUANTITY

Where does σ_r come from? The input-rate prior

The kid-level intercept $\xi_i = \log r_i + \log \alpha_i$ has total variance σ_ξ^2 that the data *does* identify. But the split between input rate (σ_r^2) and efficiency (σ_α^2) is not separately identified within the model:

$$\sigma_\xi^2 = \sigma_r^2 + \sigma_\alpha^2 \Rightarrow \pi_\alpha = 1 - \frac{\sigma_r^2}{\sigma_\xi^2}$$

We resolve this by pinning σ_r externally — from data quantifying actual variation in child-directed input rates across families. The model then partitions the data-identified σ_ξ^2 into input and efficiency components.

The pin matters for π_α .

- Bigger $\sigma_r \Rightarrow$ more variance assigned to input $\Rightarrow$ lower π_α .
- Pinning σ_r *too low* would overstate efficiency's share; *too high* would mask it.

Strategy. Compile the empirical literature on adult-token rates per child, derive a defensible σ_r , and report a full sensitivity sweep so the headline doesn't hinge on the central pin.

Empirical estimates of input-rate variation

Central pin: $\sigma_r = 0.534$ from the Sperry within-pool.

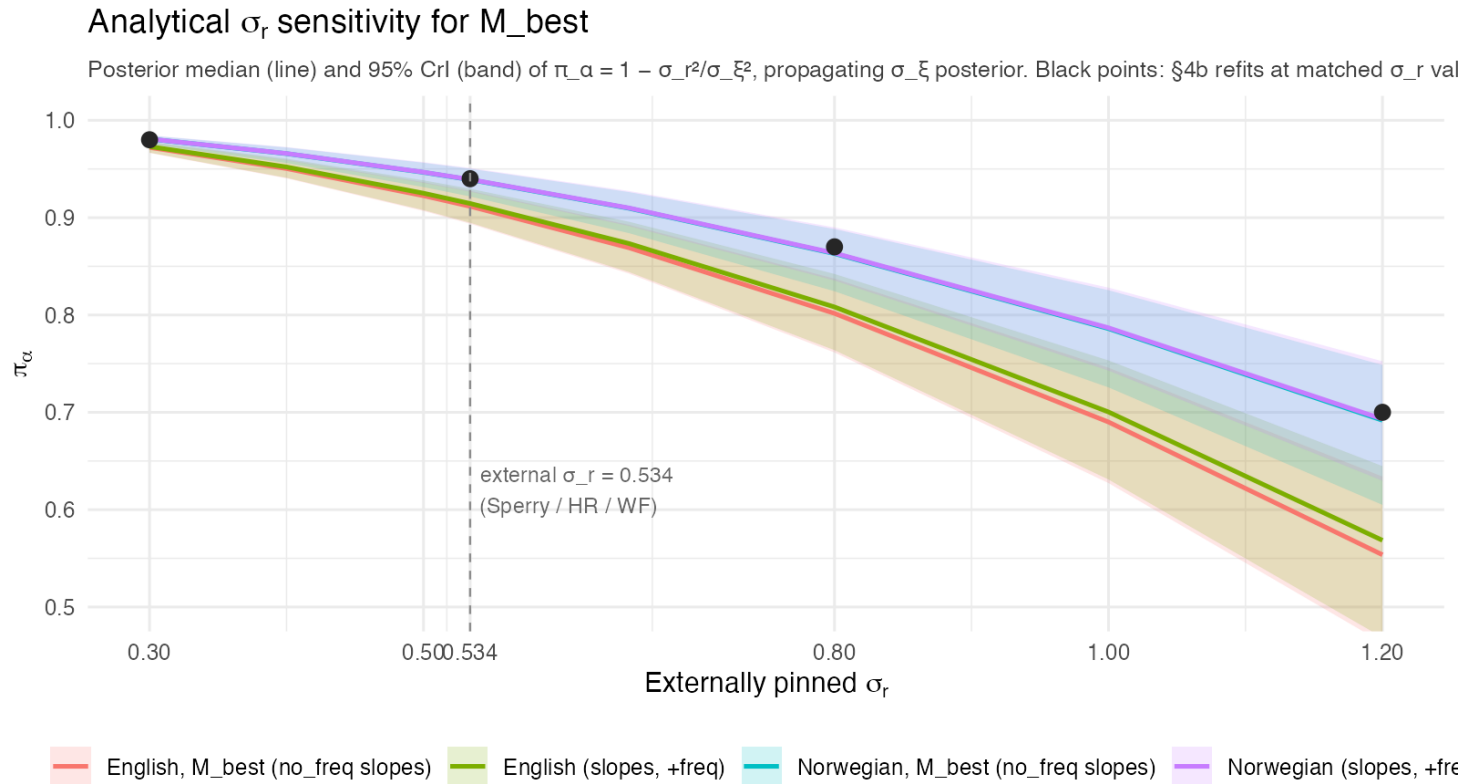
A typical Western child hears $\sim 547\text{K}$ adult tokens/month (± 1 SD: $\sim 320\text{K}$ – 960K , a $2.5\times$ spread).

The Wordbank CDI:WS sample is overwhelmingly WEIRD, so the cross-cultural 1.24 is an outer bound, not a likely value.

Tokens/month = tokens/hr $\times$ H, $H = 365$ hr/mo (12 waking hr/day $\times$ 30.44 days/mo, model convention).

Sources: Sperry et al. 2019, Hart & Risley 1995, Weisleder & Fernald 2013, Bergelson 2018/2019/2023, Bunce et al. 2024, Casillas 2020/2021, Coffey et al. 2024.

Sensitivity: π_α across plausible σ_r



Central pin $\sigma_r = 0.534$:
 $\pi_\alpha \approx 0.91 - 0.94$.

Doubled $\sigma_r = 1.2$ (cross-cultural upper bound):
 π_α still > 0.55 .

Halved $\sigma_r = 0.30$ (below any defensible single-site SD): $\pi_\alpha \approx 0.97$.

Caveat: input *quantity*, not *quality*

What we've modeled. Variation in input *quantity* — how many adult tokens per hour each child hears.

What we haven't modeled. Variation in input *quality* — diversity, contingency, conversational structure, child-directed register.

- **Quality is hard to measure at scale.** No cross-population benchmark analogous to LENA-style token rates.
- **Quality would have to be enormous to escape the finding.** A per-child age-varying quality multiplier $q_i(t) = (t/a_0)^{\gamma_i}$ is observationally equivalent to a steeper κ_i — so to absorb the observed acceleration, “quality” would need to grow $\sim 3,000\times$ for the median kid and $\sim 10^6\times$ for the 95th percentile between 12 and 30 mo. (See appendix.)

Bottom line. Quantity is too small and too well-measured to explain the variability; quality would have to be implausibly large and age-varying to substitute.

**EXTENSIONS: DIRECTLY OBSERVED
INPUT AND PROCESSING**

Model extensions: testing π_α with richer data

Wordbank longitudinal fits use *population-level* input rate priors (Sperry/Hart-Risley/Weisleder-Fernald).

Two natural follow-ups:

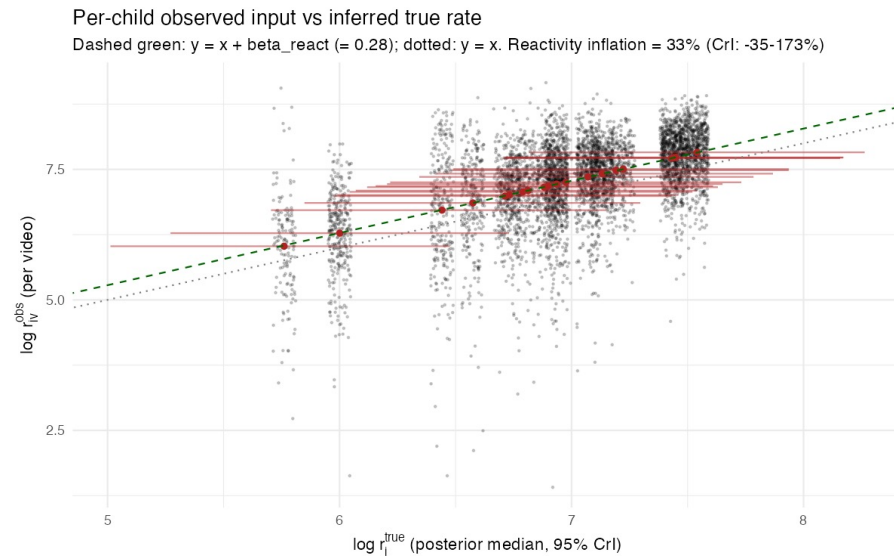
- 1. Per-child observed input:** use *measured* per-child input rate instead of a population prior. Does π_α stay ~ 0.9 when input is no longer a pooled estimate? → **BabyView, SEEDLingS.**
- 2. Per-child processing efficiency:** link CDI growth to an independent per-child readout of processing speed. Does the σ_α we estimate from CDI reflect “processing efficiency” or something else? → **Peekbank.**

Both extensions reuse the same model machinery with an added channel in the linear predictor; only the data and the per-child latents change.

Replicating with directly observed input

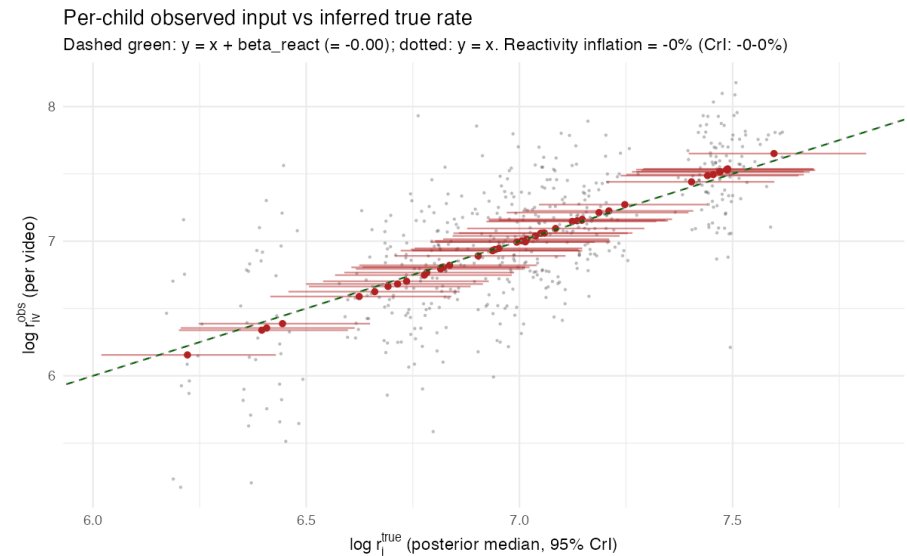
BabyView, $N = 20$

$$\pi_{\alpha} = 0.84, [0.68, 0.93]$$



SEEDLingS, $N = 44$ (comp corrected)

$$\pi_{\alpha} = 0.94, [0.89, 0.97]$$

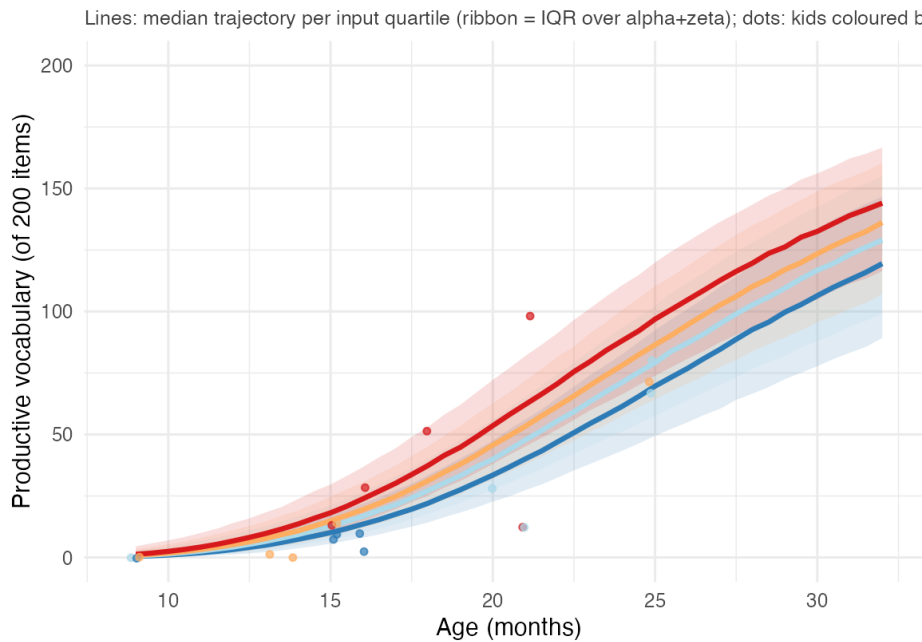


Minimal input effects

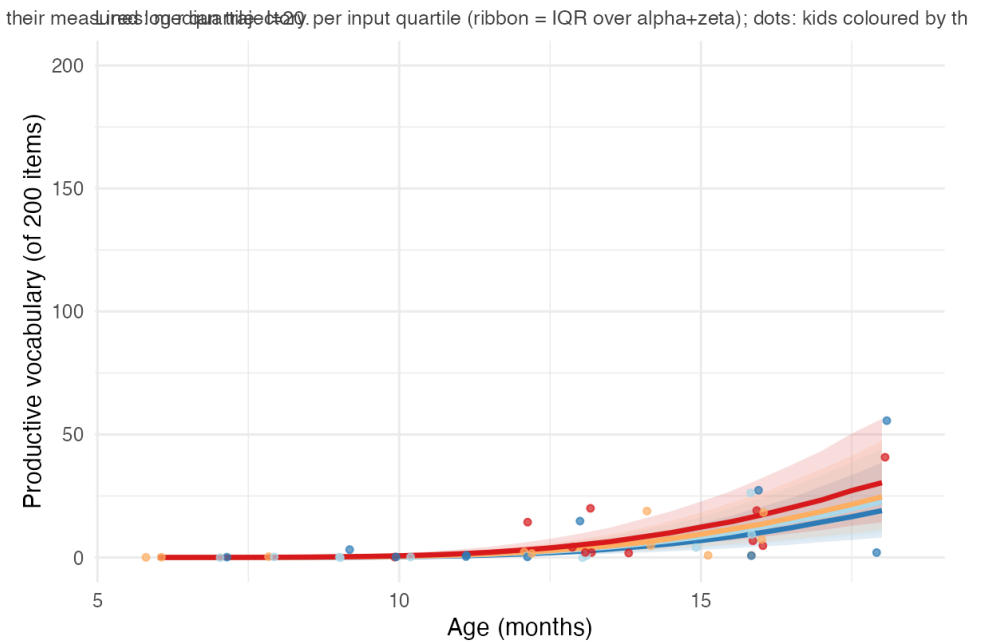
Vocabulary trajectory by input quartile

IO model (M_best, no s, no s_i). Per-kid log r (mean over recordings) split into quartiles; model trajectory conditional on each quartile's median log r.

BabyView (English)



SEEDLingS (English)



Input quartile — Q1 (lowest) — Q2 — Q3 — Q4 (highest)

Processing speed readout: Peekbank LWL

Joint model adds a per-child LWL log-RT readout coupled to $\log \alpha_i$ via γ_{rt} .

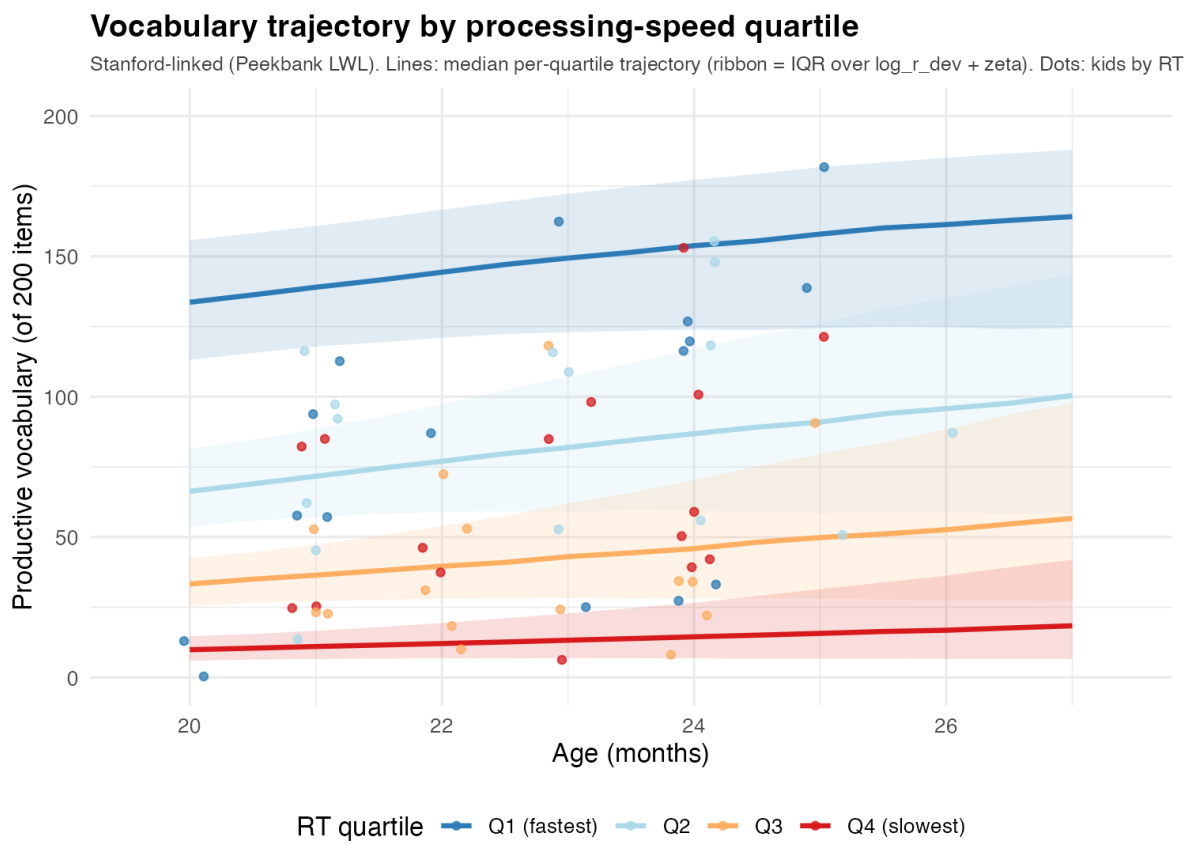
- $\gamma_{rt} = 0.084, [0.044, 0.125]$
 - vocab efficiency *level* ($\log \alpha$) and processing-speed *level* (RT intercept) are coupled: ~13% RT speedup per SD of α .
- $\pi_{\alpha} = 0.90, [0.86, 0.94]$ — replicates within Peekbank.
- $\rho(\zeta, rtslope) = +0.05, [-0.26, +0.34]$
 - but vocab acceleration and RT *maturation* are uncoupled.

One clock for the level, separate clocks for the maturation rates.

Vocabulary efficiency and processing speed coincide at any given age (the γ_{rt} link), but how fast each *changes* across development is independent.

Children don't have a single "rate of maturation" that synchronizes their vocab acceleration with their RT improvement.

Peekbank RT results



π_α across five samples

Parameter	EN M_best (I=500)	NO M_best (I=500)	BabyView (IO)	SEEDLingS (IO)	Peekbank (proc)
I (kids)	500	500	20	44	62
J (items)	671	674	200	200	200
N (obs)	1106805	2056872	16459	102800	20600
delta	10.31 [10.25, 10.37]	11.47 [11.40, 11.53]	6.42 [5.98, 6.87]	8.19 [7.82, 8.57]	2.51 [1.60, 3.38]
sigma_alpha	1.81 [1.69, 1.94]	2.05 [1.93, 2.18]	1.13 [0.82, 1.56]	1.39 [1.12, 1.74]	1.67 [1.35, 2.03]
sigma_zeta	3.81 [3.58, 4.07]	4.79 [4.49, 5.13]	2.57 [1.92, 3.39]	3.97 [3.32, 4.70]	6.42 [5.34, 7.54]
pi_alpha	0.92 [0.91, 0.93]	0.94 [0.93, 0.94]	0.83 [0.66, 0.93]	0.94 [0.89, 0.97]	0.90 [0.86, 0.94]

CONNECTING TO LLMS

Why look at LLMs?

LLMs and children both learn word meanings from input. They differ in scale and architecture, but the question of how their learning scales with input is statistically well-posed:

- For an LM and a word w , we have $P(w \text{ produced correctly})$ as a function of training tokens.
- This is a sigmoid in log-tokens with a per-word slope.
- The same statistic exists for human children: $P(w \text{ acquired})$ as a function of cumulative input.

This lets us put kids and LMs on the same axis:

1. Do kids and LMs have similar per-word acquisition slopes?
2. Does a model that fits kids' learning say anything about LMs?

Per-word sigmoids in LMs

Setup. Chang & Bergen trained four LMs (BERT, biLSTM, GPT-2, LSTM) on the same input corpus. At each training checkpoint, they probed each CDI words to get $P(\text{word produced correctly} \mid \text{checkpoint})$.

For each (LM, word) they fit a 4-parameter logistic

$$S_w(x) = \text{lower}_w + \frac{\text{upper}_w - \text{lower}_w}{1 + \exp((x - \text{mid}_w)/\text{scale}_w)}$$

where $x = \log_{10}(\text{training steps})$, yielding a per-word **scaling slope**: how fast that word's acquisition probability rises with cumulative training. After unit conversion, this is exactly analogous to our $\kappa_i \log t$ for children.

Result they reported. Across the four LMs, mean per-word slope ≈ 1 logit per nat-log training-step. Striking but ambiguous: is this the LM architecture or the input distribution?

Chang & Bergen (2022)

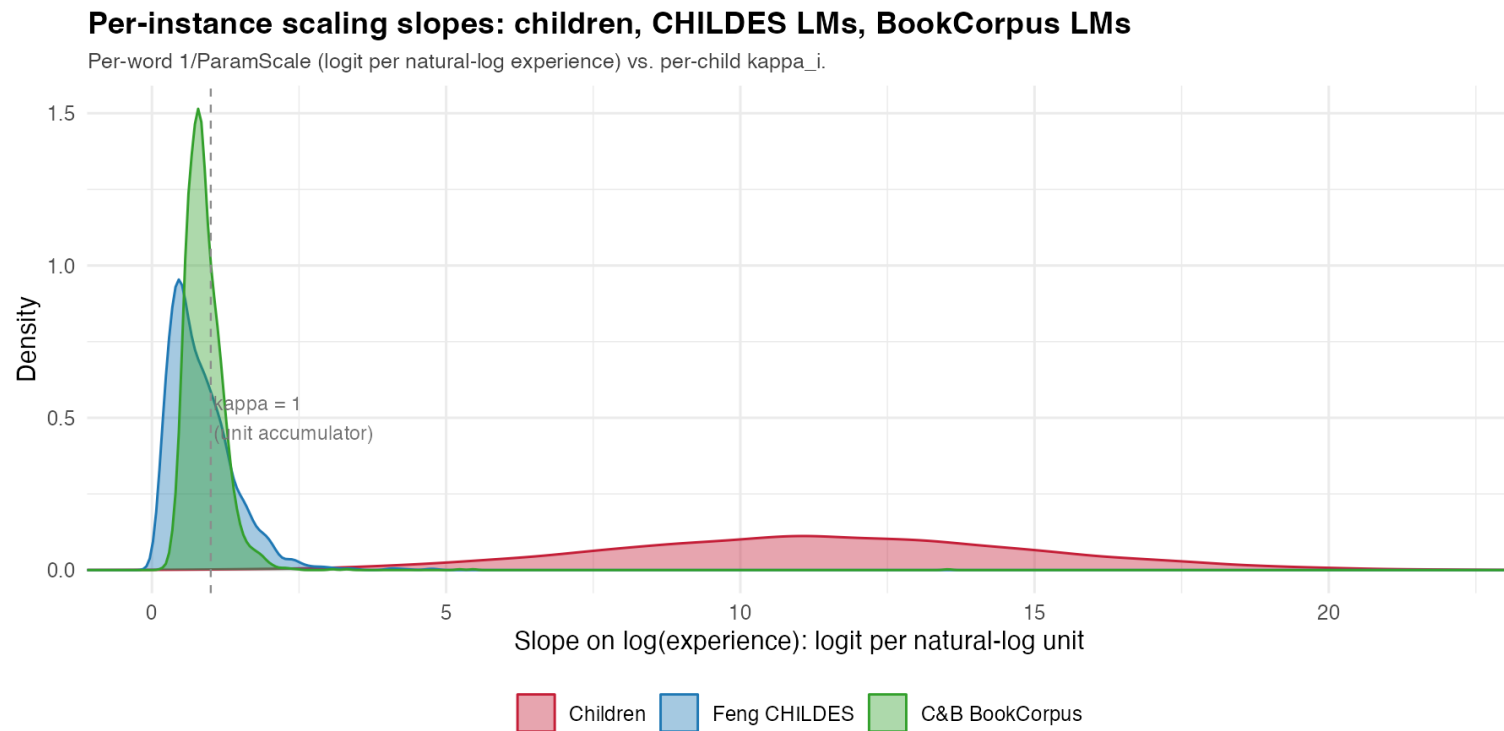
Could the gap be an input-distribution artifact?

Test: retrain GPT-2 small on CHILDES (the corpus kids actually hear, per Feng et al. 2026), keep everything else fixed, refit C&B's per-word sigmoid. If the gap is input-driven, CHILDES-trained slopes should rise toward children.

Setup:

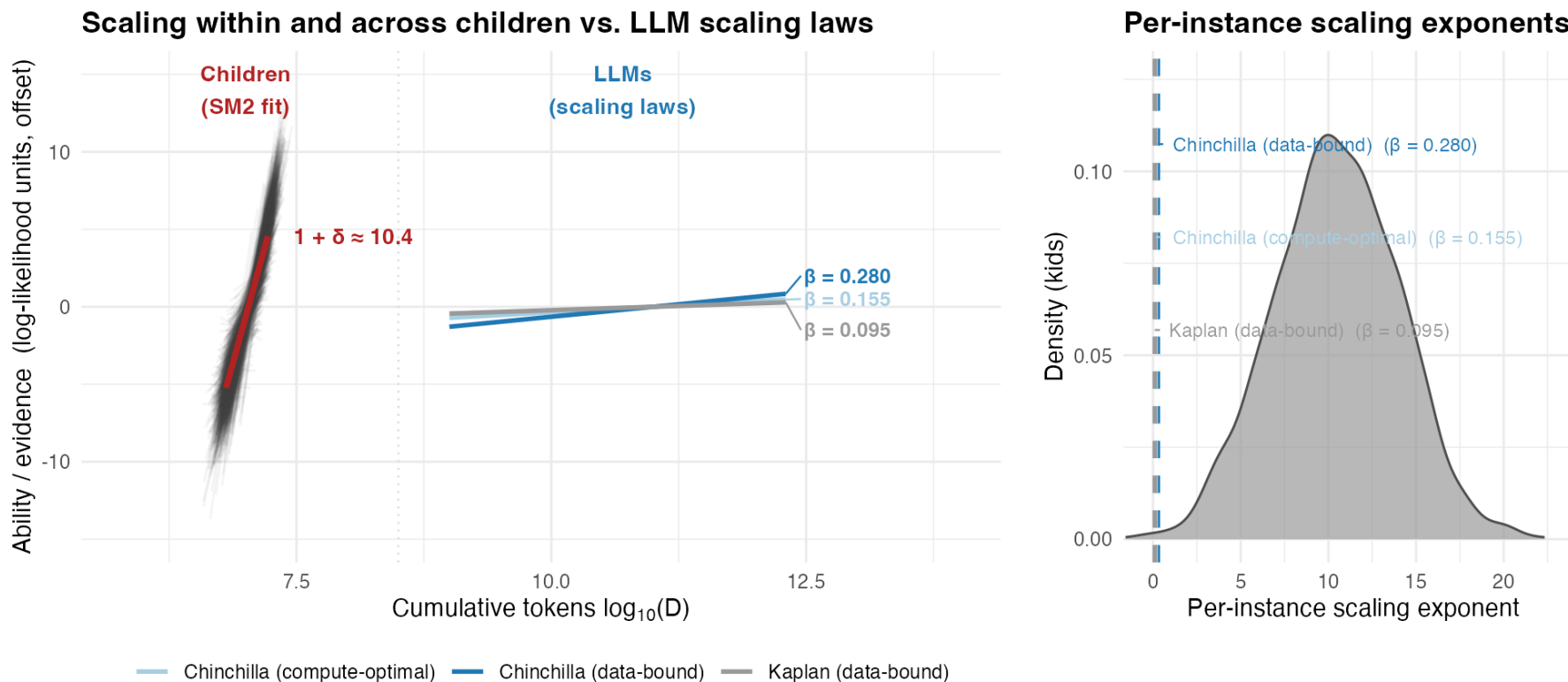
- GPT-2 small (124M params, 12L $\times$ 768d $\times$ 12h)
- CHILDES train split, $\sim$ 24.5M tokens, 20 epochs (Feng et al. §B spec)
- 3 random seeds ($\{0, 42, 123\}$) as a between-instance variance probe
- 73 log-spaced eval checkpoints; per-word mean NLL over 50 occurrences in the CHILDES validation set
- C&B 4-parameter logistic fit per (LM, word); per-word slope = $0.434 / \text{ParamScale}_w$
- 609 / 611 C&B CDI words are single tokens in the GPT2-CHILDES BPE

CDS training does not move the LM slope



The aggregate view: scaling laws

Slopes are the structurally meaningful comparison; y-axis offset is anchored for visual clarity. Kid D range comes from per-kid r_i .



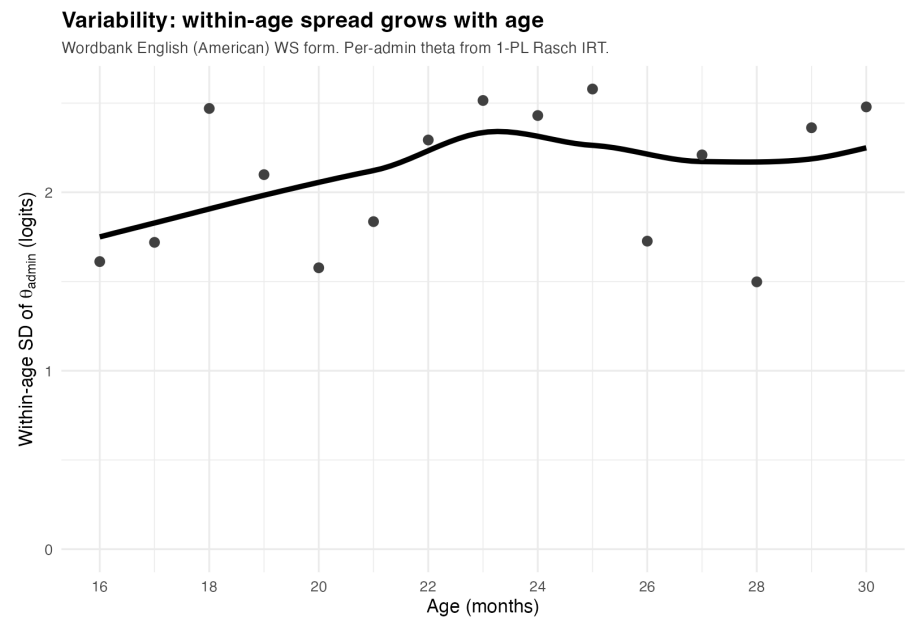
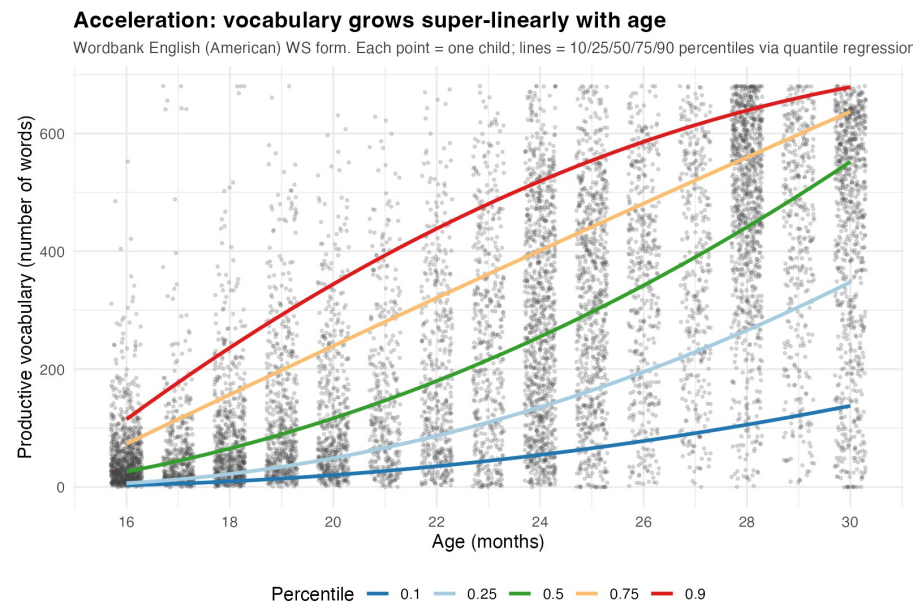
SM2 fit 'long_slopes': median $1 + \delta = 10.39$ [10.24, 10.55]; $\sigma_{\zeta} = 3.48$ [3.15, 3.87]; $\mu_r = 7.34$, $\sigma_r = 0.53$, $a_0 = 19$. Chinchilla and Kaplan exponents from Hoffmann et al. 2022 / Kaplan et al. 2020.

SYNTHESIS

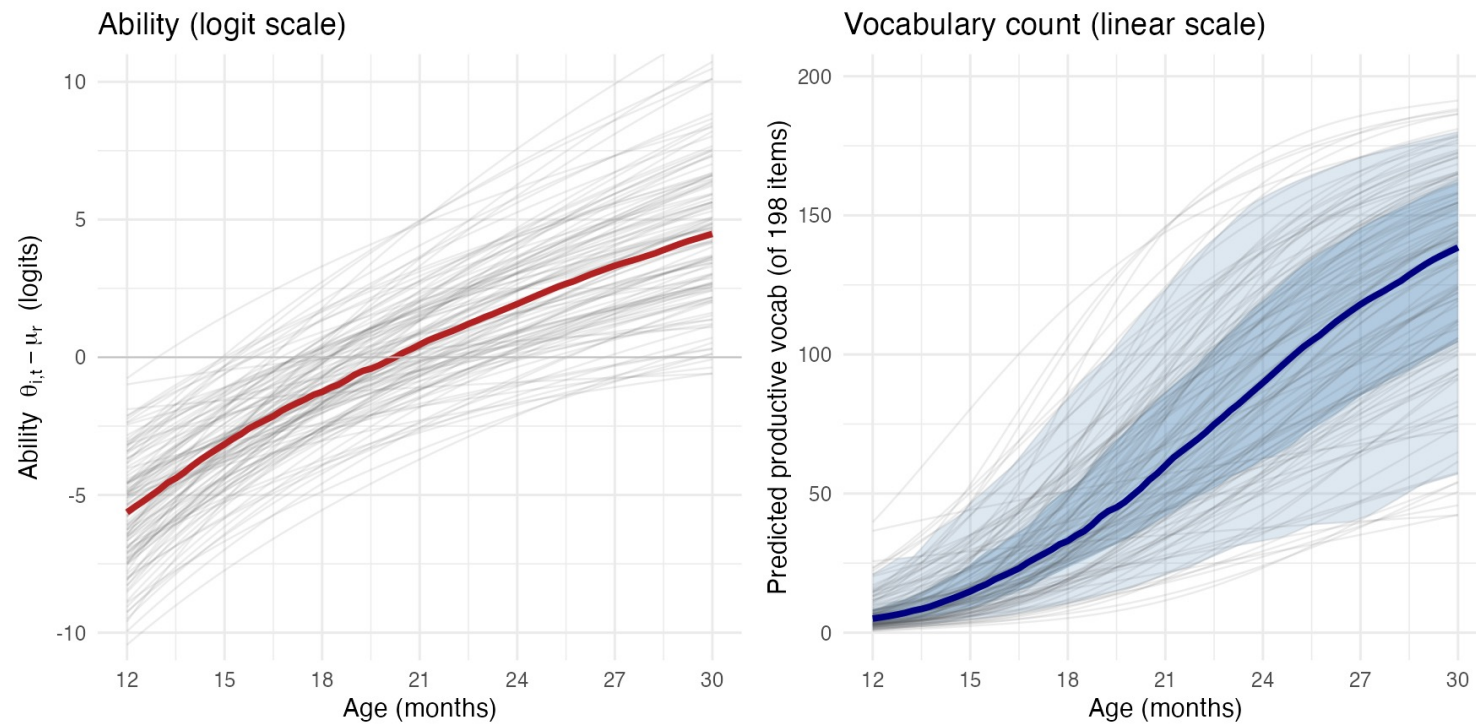
Technical summary

- **Acceleration is real and population-level.** $\kappa_{\text{pop}} \approx 10$, confidently above the unit-accumulator boundary $\kappa_{\text{pop}} = 1$. The first prior model to free κ_{pop} (Hidaka rate-change) was structurally right.
- **Individual variability in acceleration is real too.** $\sigma_z > 0$ and $\sigma_s > 0$: children differ not just in baseline ability, but in *how their trajectories accelerate* and *when*. No prior accumulator model frees these.
- **Input quantity is a small share of between-child variance.** $\pi_\alpha \approx 0.91$ across 5 samples. Aligns with the meta-analytic $R^2 = 0.04\text{--}0.07$ of Coffey & Snedeker (2026).
- **LLMs are structurally different.** Same per-word sigmoid statistic gives $\kappa \sim 1$ for LMs vs. ~ 10 for children — a 10× gap with near-zero between-instance variance in LMs.

Two signatures, twin constraints on theory

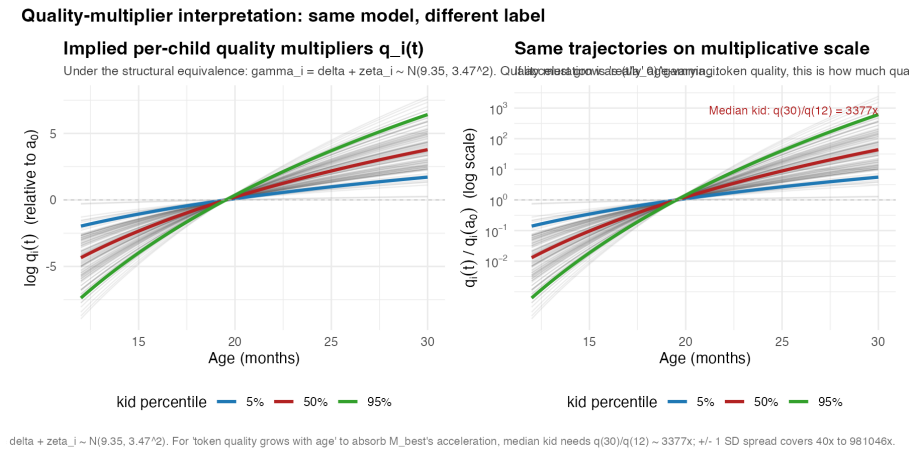


Appendix: per-child trajectories from the best fit



120 kids sampled from the best-fitting model's posterior. **Left:** latent ability θ (logit). **Right:** predicted vocabulary. Concave-up shape = super-linear signature; fan width = between-child variability.

Appendix: σ_r sensitivity from earlier cross-sectional fits



Reviewer reflex. “Maybe kids’ tokens differ in *quality*, and that’s what looks like acceleration.”

Structural answer. Per-child age-varying quality $q_i(t) = (t/a_0)^{\gamma_i}$ on input is observationally equivalent to $\gamma_i = \kappa_i - 1$.

Implication. Quality cannot escape acceleration — it *relabels* it. For the median kid, “quality” would have to grow $\sim 3,000\times$ from age 12 to 30 mo. For the 95th percentile kid, $\sim 10^6\times$. Implausible as a substantive quality story.

Appendix: McMurray (2007) in our notation

Original. Each word w has threshold $\tau_w \sim g(\tau)$; deterministic step at $t \geq \tau_w$. Vocabulary $L(t) = G(t)$ (CDF of g). Acceleration requires $g'(\tau) > 0$ over the productive window.

In our notation:

$$\eta_{ij}(t) = \log t - \log \tau_j = \kappa \log t - \delta_j \text{ with } \kappa \equiv 1, \delta_j = \log \tau_j.$$

Appendix: Hidaka (2013) in our notation

Original. Three families: cumulative (N -accumulator at constant rate, AoA $\sim \Gamma$), rate-change (1-accumulator at rate $\delta \cdot t^D$, AoA $\sim$ Weibull), and both. **Rate-change:** ability $\propto t^{D+1}$; $D > 0$ produces super-linear scaling.

In our notation (rate-change form):

$$\eta_{ij}(t) = (D + 1)\log t - \delta_j \text{ with } \kappa_i \equiv D + 1.$$

Hidaka's $D = \kappa_{\text{pop}} - 1$ **in our notation.** His $D > 0$ is our $\kappa_{\text{pop}} > 1$.

Appendix: Mollica & Piantadosi (2017) in our notation

Original. Per-word: k_w effective learning instances arrive as Poisson at rate λ_w ; word acquired after k_w events.

$AoA_w \sim \Gamma(k_w, \lambda_w)$. For large k_w , Gamma CDF approximates Gaussian (mean k_w/λ_w) and then a ~ 1.7 -scale logistic.

In our notation:

$$\eta_{ij}(t) = \log t - \delta_j \text{ with } \kappa_i \equiv 1, \delta_j \approx \log(k_j/\lambda_j).$$

Appendix: Kachergis et al. (2021) in our notation

Original. 1-PL Rasch IRT with per-child latent ability:
 $\text{logit } P(y_{ij} = 1) = \theta_i + d_j + \beta_a a_i + \beta_t \log(p_j H a_i)$. Per-child θ_i free; d_j per-word free; shared age slope β_a across all kids.

In our notation:

$$\eta_{ij} = (\log r_i + \log \alpha_i) + \kappa \log a_i - \delta_j \text{ with } \kappa \\ \equiv \text{shared population slope.}$$

First prior model to free σ_α . But σ_κ is still zero — every kid has the same scaling slope.